QUESTION 1 [20 marks]

a. **(10 marks)** The circuit in Figure 1 is a diode-based protection circuit for a portable TV. The red LED will glow to indicate "FAULT — battery reversed" in case of wrong connection to a DC power supply (reversed polarity). With the correct polarity, the green LED will glow to indicate "OK — normal operation".

The voltage drop across the 1N5400 diode and LEDs under forward bias is 2.1 volts. The TV is represented as a 200 Ω resistance.

- **i.** **(5 marks)** Calculate the current through the TV and green LED when the circuit is in normal operation.
- **ii.** **(3 marks)** Calculate the current through the red LED when the circuit is in FAULT mode.
- **iii.** **(2 marks)** Describe how the 1N5400 diode can protect the TV when the polarity is reversed.

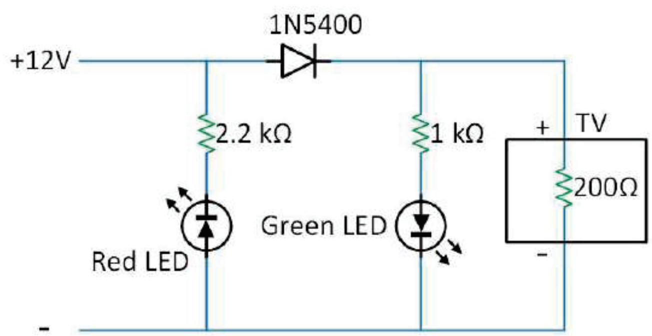

Figure 1

b. **(10 marks)** Find the Norton equivalent of the circuit shown in Figure 2 from terminals *a*-*b*.

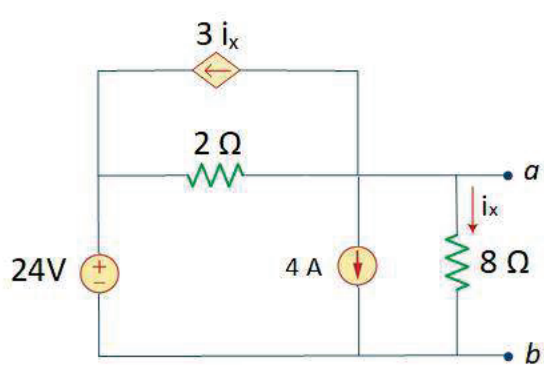

Figure 2

QUESTION 2 [20 marks]

a. **(8 marks)** The flash unit of a camera is shown in Figure 3, where the flashlight is represented as a 12 Ω resistance. The capacitor is charged when the switch is at position *a*. This charge then provides a burst of power to the flashlight when taking a picture, that is, when the switch is moved to position *b*.

- **i.** **(5 marks)** Assume that the switch has been at position *a* for a long time. The switch is moved to position *b* to take a picture and the flash fires for 1.2 ms. Find the voltage v_0 across the capacitor once the picture has been taken (at t=1.2 ms).
- **ii.** **(3 marks)** Once the picture is taken at t=1.2 ms, the switch is moved back to position *a*. How long will it take for the capacitor to fully recharge before taking the next picture?

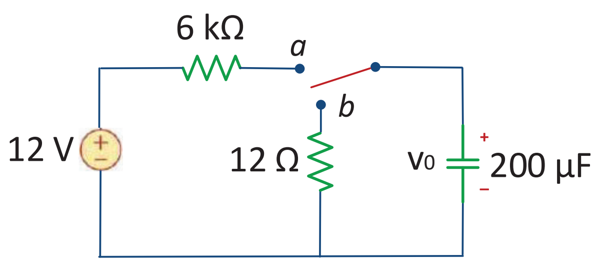

Figure 3

b. **(12 marks)** Switch 1 (SW1) in Figure 4 was in position *a* for a long time before moving to position *b* at t=0. After 0.1 seconds, the voltage source is disconnected from the circuit by opening Switch 2 (SW2).

- **i.** **(6 marks)** Derive the expression for $i_0(t)$ for $0 \leq t < 0.1$ s.
- **ii.** **(4 marks)** Derive the expression for $i_0(t)$ for $t \geq 0.1$ s.
- **iii.** **(2 marks)** Calculate the value of i_0 at $t = 0.2$s.

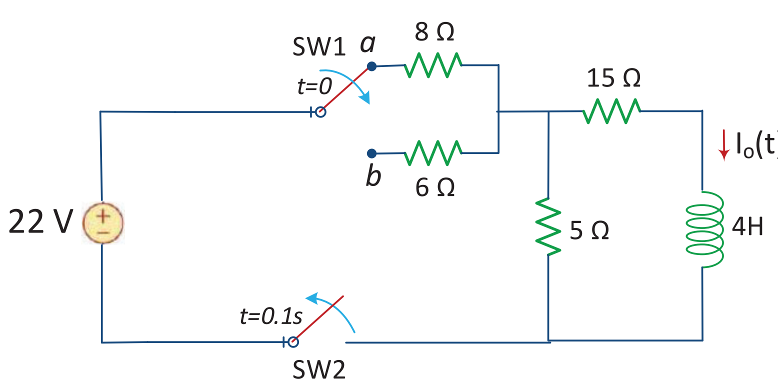

Figure 4

QUESTION 3 [22 marks]

a. (10 marks) Resistance temperature detectors (RTDs), are sensors used to measure temperature based on the resistance changes. The most common type (PT100) has a resistance of 100 Ω at 0°C and 138.4 Ω at 100°C. The circuit shown in Figure 5 is used to convert the information from the RTD into something meaningful for an analog-to-digital converter (ADC).

i. (8 marks) Calculate the voltage v_0 as a function of v_{in} when the RTD has a value of 130 Ω.

ii. (2 marks) How large can be the value of v_{in} before the amplifier saturates?

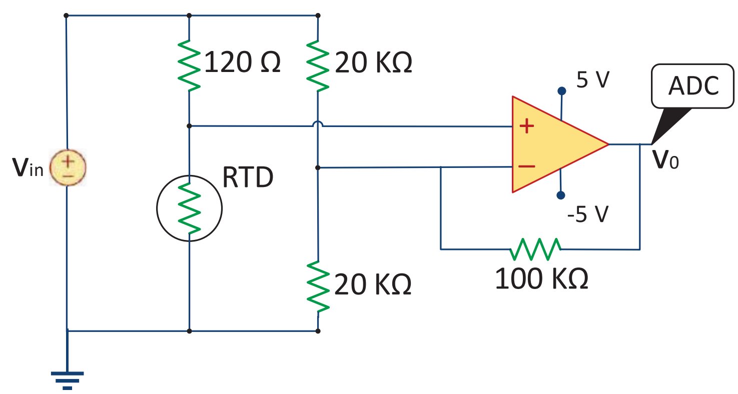

Figure 5

b. (12 marks) Calculate the output voltage $v_o(t)$ in the circuit shown in Figure 6.

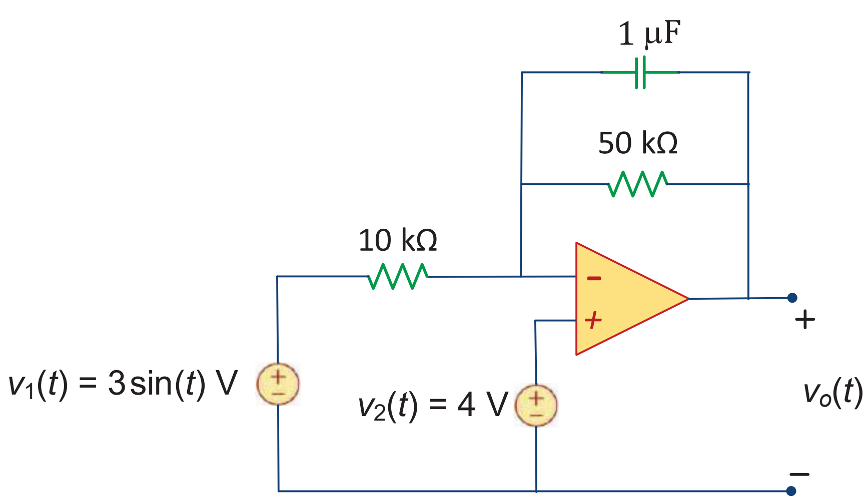

Figure 6

QUESTION 4 [20 marks]

a. **(8 marks)** A regular household system of a single-phase three-wire circuit allows the operation of both 120 V and 240 V, 60 Hz appliances. The household circuit is shown in Figure 7. Calculate the current through each of the appliances (i_{lamp}, i_{ref} and i_{range}).

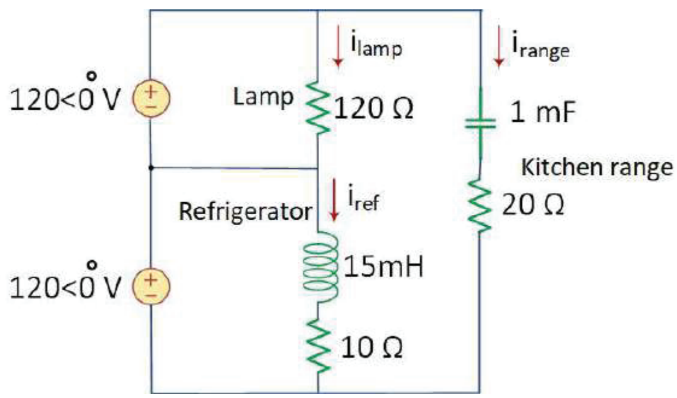

Figure 7

b. **(12 marks)** Calculate $v_0(t)$ in the circuit of Figure 8.

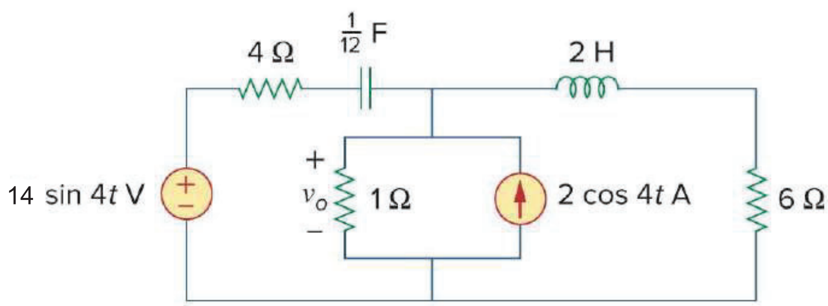

Figure 8

QUESTION 5 [18 marks]

a. (9 marks) A coupling capacitor is used to block the DC current from an amplifier as shown in Figure 9(a). The amplifier and the capacitor act as the source, while the speaker is the load as in Figure 9(b).

i. **(5 marks)** Calculate the value of R_L and the source frequency such that the circuit can deliver maximum average power to the speaker.

ii. **(4 marks)** If $|V_{in}| = 4.6\ V_{rms}$, how much power is delivered to the speaker at that frequency?

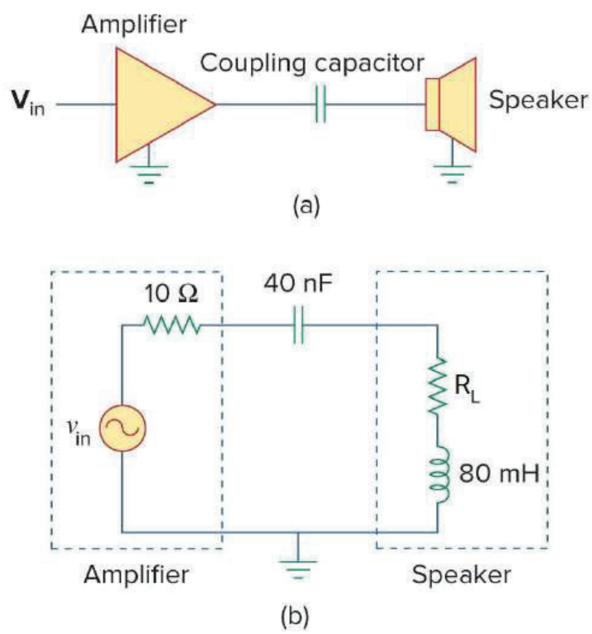

Figure 9

b. (9 marks) Given the circuit in Figure 10, find the average power absorbed by the 10-Ω resistor.

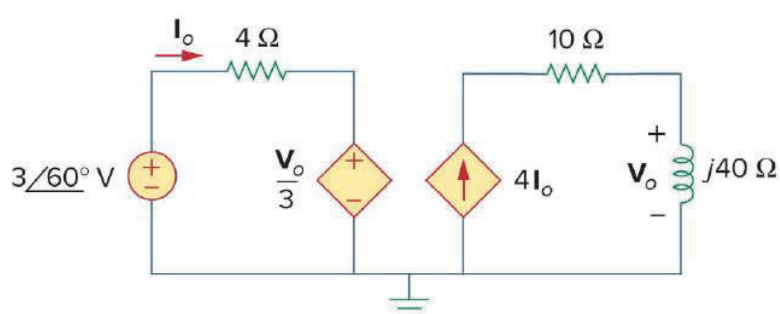

Figure 10